Figure 15.19 Checking whether the point of intersection of the two lines falls within both segments.

In other words, the coordinates of point P can be expressed in terms of both vectors, and the following equation must hold:

$$\mathbf{B} + h\mathbf{v} = \mathbf{D} + g\mathbf{w} \quad (1)$$

Notice that B and D can also be considered vectors: in particular, the vectors that go from the origin to those two points.

We had mentioned we would define a new vector u: well, now it's the time, we consider

$$\mathbf{u} = (-v_y, v_x) = (A_y - B_y, B_x - A_x)$$

This vector has a special property: its dot product with vector v is zero:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u} = v_x v_y - v_y v_x = 0$$

Going back to equation (1) above, we can multiply both side by u, which is not null (as long as A and B are distinct points, which we assume as hypothesis, because vertices can't be assigned to the same point). Doing so we get: